



JC4002 Knowledge Representation

Day 3: Resolution with Horn Clause

Dr Ferdian Jovan

ferdian.jovan@abdn.ac.uk

Dr Aiden Durrant

aiden.durrant@abdn.ac.uk

Dr Jiangtian Nie

jiangtian.nie@abdn.ac.uk

Day 3

Resolution with Horn Clause

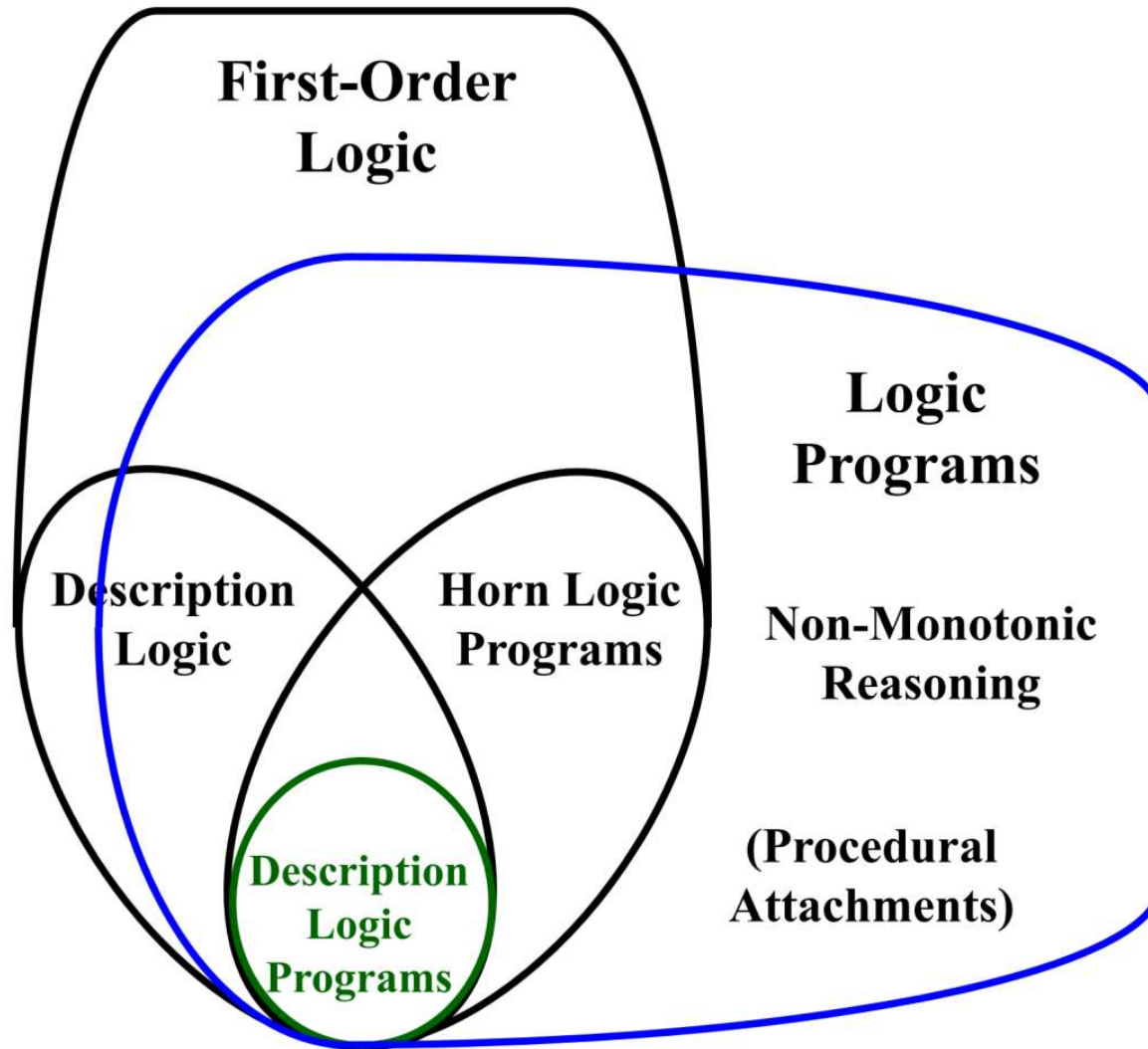
1. Horn Clauses
2. Resolution with Horn Clause
3. Backward-chaining Procedure
4. Forward-chaining Procedure
5. Next Steps

FOL vs Horn Clause

- FOL is very expressive and powerful, but we have seen some of the problems that come with resolution in such language (undecidability, intractability of first-order entailment, computational cost, bothersome syntax, etc.)
- We have to limit this to a particular subset of FOL that remains expressive for most problems, but becomes much more manageable for automated resolution procedures
- Horn clauses are one of the options that we have to do this
- A clause in resolution-based systems is a **disjunction of literals** (atomic formula or its negation), such as: $(rain \vee sleet \vee snow)$

Incomplete knowledge, there is rain or sleet or snow outside, but we do not know which

FOL vs Other Logics



Horn Clauses

- Named after the logician Alfred Horn, who first provided evidence of their significance in 1951

- A Horn clause is a clause with **at most one positive literal**

- $(\neg \text{child} \vee \neg \text{male} \vee \text{boy})$

Either someone is not a child, or not a male, or a boy

- Which by definition can be represented as an **implication**:

- $(\text{child} \wedge \text{male} \rightarrow \text{boy})$

- If someone is a child and a male then then that someone is a boy

- More generally

$$(\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n \vee q)$$

- Then

$$((p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q)$$

Types of Horn Clause

- Definite (or positive) clause: Horn clause with exactly one positive literal
- Fact: definite clause with no negative literals and without variables
- Goal (or negative) clause: Horn clause with no positive literal¹

Type	Disjunction	Implication	Read as
Definite clause	$\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n \vee q$	$p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow q$	$p_1 \dots p_n$ holds, then q also holds
Fact	q	$true \rightarrow q$	assume that q holds
Goal clause	$\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n$	$p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow$ <i>false</i>	show that $p_1 \dots$ p_n all hold

¹Empty clause ([]) is equivalent to false and thus is a goal clause

Exercise

Match the right column with the left

(D) Definite clause

(F) Fact

(G) Goal clause

(N) Not Horn clause

() $P(x)$

() $\forall x P(x)$

() $\exists x P(x)$

() $true \rightarrow Q$

() $P(x) \vee Q$

() $P(x) \wedge Q$

() $P(x) \wedge Q \rightarrow R(x)$

() $P(x) \wedge Q \rightarrow false$

() $\neg P(x) \vee \neg Q \vee R(x)$

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() $P(x) \wedge Q \wedge \neg R(x) \rightarrow S$

() *Lecturer(Ferdì)*

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Day 3

Resolution with Horn Clause

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Resolution

- ❑ Takes two Horn clauses and produces another clause
- ❑ The new clause is implied by the two previous clauses
- ❑ Removes complementary literals from the new clause
- ❑ Terminates when the empty clause is produced ($[]$)
- ❑ Behind the scenes implementation of this uses something called Selective Linear Definite (SLD) clause resolution (more information in chapter 5.2 and 5.3 of our textbook)

Activity 1

In pairs: try to prove

“ $(p \rightarrow q)$ and p entails q ”

Discuss with other pairs your answer

Resolution Simple Example

For the premise $(p \rightarrow q)$ and p we want to prove q .

Through resolution:

- | | |
|--------------------|--------------|
| 1. $\{\neg p, q\}$ | Premise |
| 2. $\{p\}$ | Premise |
| 3. $\{\neg q\}$ | Negated Goal |
| 4. $\{q\}$ | 1, 2 |
| 5. $\{\}$ | 3, 4 |

Resolution Another Example

Horn Clause KB (another format): $[\neg p \vee \neg q \vee r]$ $[p]$ $[q]$ $[\neg r]$

Resolution Another Example

Horn Clause KB (another format):

$[\neg p \vee \neg q \vee r]$	$[p]$	$[q]$	$[\neg r]$
$[\neg p \vee \neg q \vee r]$	$[p]$	$[q]$	$[\neg r]$

$\underbrace{\hspace{10em}}$

Resolution Another Example

Horn Clause KB (another format):

$$\begin{array}{l} [\neg p \vee \neg q \vee r] \quad [p] \quad [q] \quad [\neg r] \\ [\neg p \vee \neg q \vee r] \quad [p] \quad [q] \quad [\neg r] \\ \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ [\neg q \vee r] \quad [q] \quad [\neg r] \\ \quad \quad \quad \uparrow \quad \quad \quad \uparrow \end{array}$$

Resolution Another Example

Horn Clause KB (another format):

$[\neg p \vee \neg q \vee r]$	$[p]$	$[q]$	$[\neg r]$
$[\neg p \vee \neg q \vee r]$	$[p]$	$[q]$	$[\neg r]$
			
$[\neg q \vee r]$	$[q]$	$[\neg r]$	
			
$[r]$	$[\neg r]$		
			

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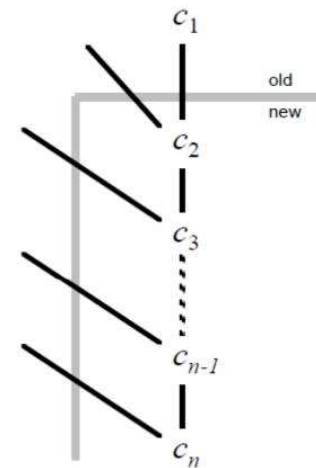
$$[\neg p \vee \neg q \vee r] \quad [p] \quad [q] \quad [\neg r]$$
$$[\neg q \vee r] \quad [q] \quad [\neg r]$$
$$[r] \quad [\neg r]$$

[]

Further Restricting Resolution

Can also change derivations such that each derived clause is a resolvent of the previous derived one (negative) and some positive clause in the original set of clauses

- Since each derived clause is **negative**, one parent must be positive (and so from the original set) and one parent must be negative
- Chain backwards from the final negative clause until both parents are from the original set of clauses
- Eliminate all other clauses not in this direct path



Restricting Resolution Example

Horn Clause

KB

FirstGrade

FirstGrade $\rightarrow$ *Child*

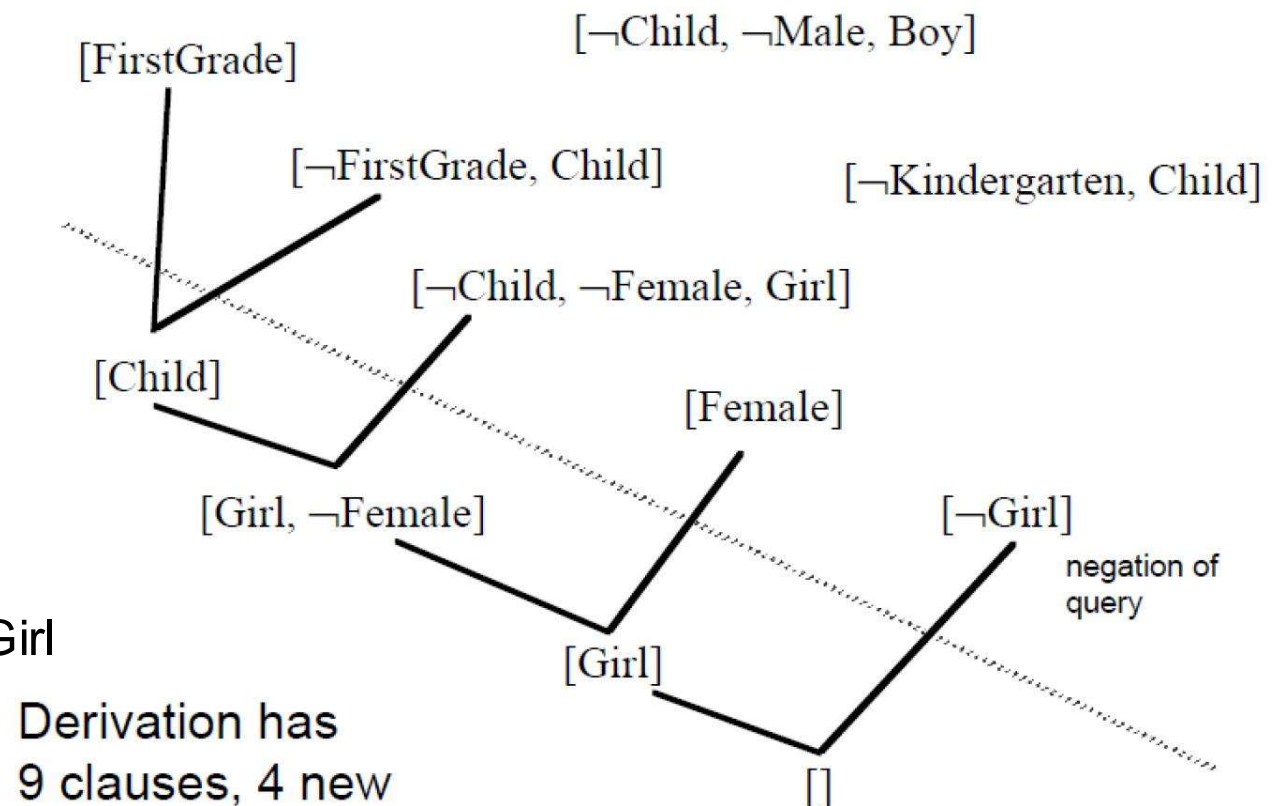
Child $\wedge$ *Male* $\rightarrow$ *Boy*

Kindergarten $\rightarrow$ *Child*

Child $\wedge$ *Female* $\rightarrow$ *Girl*

Female

Show $KB \cup \{Girl\}$ i.e. $KB \models Girl$



Derivation has
9 clauses, 4 new

SLD Resolution

An **SLD-derivation** of a clause c from a set of clauses S is a sequence of clauses $c_1, c_2, \dots, c_n$ such that $c_n = c$ and

1. $c_1 \in S$
2. c_{i+1} is a resolvent of c_i and a clause in S

Write $S \xrightarrow{\text{SLD}} c$ **(S)**electd **(L)**inear **(D)**efinite clauses

SLD-derivation is just a special form of derivation where we leave out the element of S (except c_1)

In general, we can't restrict ourselves to just using SLD-Resolution

- Proof: $S = \{[p, q], [p, \neg q], [\neg p, q], [\neg p, \neg q]\}$. Then $S \rightarrow []$.
- Need to resolve some $[p]$ and $[\neg p]$ to get $[]$
- But S does not contain any unit clauses
- So we will need to derive both $[p]$ and $[\neg p]$ and then resolve them together.

Completeness of SLD for Horn Clause

However, for Horn clauses, we can restrict ourselves to **SLD-Resolution**

Theorem:

SLD-Resolution is refutation complete for Horn clauses:

$$H \longrightarrow [] \text{ iff } H \xrightarrow{\text{SLD}} []$$

This will considerably simplify the search for derivations

Note: in Horn version of SLD-Resolution, each clause in the $c_1, c_2, \dots, c_n$ will be negative. So clauses H must contain **at least one negative clause** for c_1 , and this will be the only negative clause of H used.

Typical case:

- KB is a collection of positive Horn clauses (i.e. implication clauses)
- Negation of query is the negative clause

SLD Version of Previous Example

Horn Clause

$[\neg \text{Child}, \neg \text{Female}, \text{Girl}]$

$\neg \text{Girl}$

SLD derivation

KB

FirstGrade

FirstGrade $\rightarrow$ *Child*

Child $\wedge$ *Male* $\rightarrow$ *Boy*

Kindergarten $\rightarrow$ *Child*

Child $\wedge$ *Female* $\rightarrow$ *Girl*

Female

Show $KB \cup \{\text{Girl}\}$ i.e. $KB \models \text{Girl}$

[Female]

$[\neg \text{FirstGrade}, \text{Child}]$

[FirstGrade]

$[\neg \text{Child}, \neg \text{Female}]$

$[\neg \text{Child}]$

$[\neg \text{FirstGrade}]$

[]

$[\neg \text{Girl}]$

$[\neg \text{Child}, \neg \text{Female}]$

$[\neg \text{Child}]$

$[\neg \text{FirstGrade}]$

[]

Day 3

Resolution with Horn Clause

1. Horn Clauses
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Backward Chaining

$Solve[q_1, q_2, \dots, q_n] =$
If $n = 0$ then return **YES** /* empty clause found */
for each $d \in KB$ do
 If $d = [q_1 \vee \neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_m]$ /* match first q */
 and $Solve[p_1, p_2, \dots, p_m, q_2, \dots, q_n]$ /* call itself recursively */
 then return **YES**
end for; /* no clause can be found to eliminate q */
return **NO**

- ☐ Depth-first because it attempts p_i before trying q_i
- ☐ Left to right because it tries q_i in order with $i = 1, 2, 3, \dots$
- ☐ Backward-chaining because it searches from a goal q , working backwards to the facts in the KB

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Forward Chaining

Procedure to determine if Horn clause $KB \models q$ by marking literals as solved

1. If q is marked as solved, then return YES
 2. Is there a $\{p_1 \vee \neg p_2, \vee \dots \vee \neg p_n\} \in KB$ such that $p_2, \dots, p_n$ are marked as solved, but the positive literal p_1 is not?
 - No: return **NO**
 - Yes: mark p_1 as solved and go to (1)
-
- ❑ Only literals in the KB can be marked, so at most we are looking at a linear number of iterations
 - ❑ Not goal directed

Forward Chaining

Horn Clause

KB

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FirstGrade $\rightarrow$ *Child*

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Female

Show $KB \cup \{Girl\}$ i.e. $KB \models Girl$

Marks:

1. *FirstGrade* (all the negative literals in the clause are marked)
2. *Child* (remember *FirstGrade* $\rightarrow$ *Child* is logically equivalent to $\neg FirstGrade \vee Child$)
3. *Female*
4. *Girl*

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Why Horn Clause?

- ☐ Resolution of two Horn clauses always result in a new Horn clause (as we have seen in the examples)
- ☐ Resolution of a goal clause and a definite clause is always a goal clause
- ☐ Horn clauses are more computationally tractable than other clauses
- ☐ Prolog is based on resolving Horn clauses

Summary

- Introduction to Horn clauses
- The different types of Horn clauses
- How resolution can be done with Horn clauses
- Back- and Forward-chaining procedures for resolution with Horn clauses

Until the next session:

- Install SWI-Prolog to your computer.